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THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITIES

ARUSHA REGIONAL MOCK EXAMINATION

FORM TWO

043 **MATHEMATICS**

TIME: 2:30HRS **MAY 2026**

INSTRUCTIONS

- Answer All Questions

1. (a) Explain three importances of mathematics in daily life and give one real – life example where mathematics is used in your community.
   (b)(i) Represent $-4 \leq x < 5$ on a number line, where x is a real number

   (ii) A submarine which is 200m below sea level targets an earoplane 200m above the sea level. What is the distance between them?

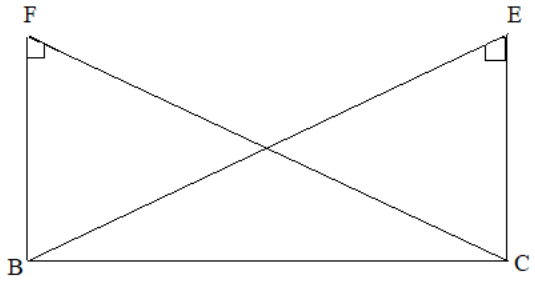
2. (a) A fraction was converted to decimal and the result was 0.441441441…… Find the fraction and express it as a ratio.
   (b) Two pipes A and B are used to fill a water tank. Pipe A can fill the tank in 6 hours while pipe B can fill the same tank in 4hours. If both pipes are opened at the same time, how long will it take to fill the tank?
   (c) A traveller exchange USD 500 to Tsh, at a rate of 1usd = 2,500/= Tsh. How many Tsh did the traveller receive?

3. (a)(i)If x varies directly as y and inversely as z and x = 8, when y = 12 and z = 6, find the value of x when y = 16 and z = 4.
   (ii) A contractor is estimating the building materials for a classroom. He needs 49 bags of cements @19700/=, two trips of sand @56500/= and 2 backets of paint @147500/=. Estimate the total cost of the materials required.
   (b) Write the number 0.002758 correct to;
   (i) 2 significant figures
   (ii) 3 decimal places

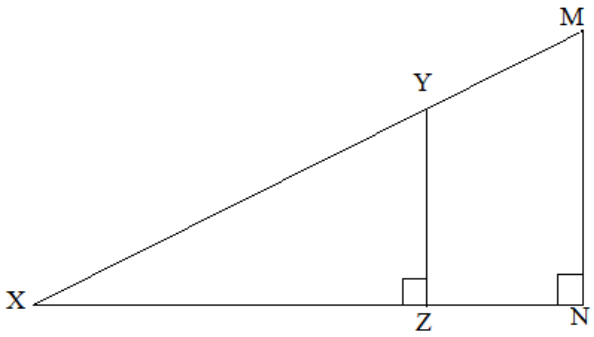
4. (a) (i) In the following figure $\overline{BF} = \overline{CE}$; prove that $\overline{CF} = \overline{BE}$

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(ii) In the following figure, find the ratio of $\overline{MY}$ *to* $\overline{MN}$.



(b) (i) If $a * b = a^2 - b$; find the value of y given that $4 * (2 * y) = 4$.

(ii) A certain formula of Physics is expressed as follows

$T = 2\pi\sqrt{\frac{l}{g}}$. Express g in terms of other letters .

5. (a) (i) Solve for the value of x if $|6 - x| = 1$

(ii) A farmer bought chickens and goats for his farm. He bought a total of ten animals. The total cost was 60,000/= with each chicken cost 4,000/= and each goat costing 8,000/=. How many chickens and goats did the farmer buy.

(b) (i) Factorize $a^2 - b^2$; and use the result to evaluate $(85.18)^2 - (14.82)^2$ without using calculator

(ii) What must be added to $5x^2 - 3x$ to make if a perfect square expression.

6. (a) (i) Solve for y given that $\frac{2y}{5} - \frac{y-2}{3} = 5$

(ii) Use quadratic formula to solve $-4x^2 + 8x - 3 = 0$

(b)(i) Rationalize the denominator of $\frac{2\sqrt{7}}{\sqrt{10}-\sqrt{7}}$

(ii) Solve for x if $5^x + 5^{(x+1)} = 30$

7. (a) Write each of the following numbers in standard from

(i) 5670000

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(ii) 0.00024

(b)(i) Without using calculators find the value of $2log5 + log36 - log\ 9$.

(ii) Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find the value of $\log_3\left(\sqrt[3]{60}\right)$, correct to 3 significant figures.

8. (a) Given the universal set $\mu = \{\text{Natural numbers less than } 12\}$

A = {Prime numbers }

B = {Odd numbers }

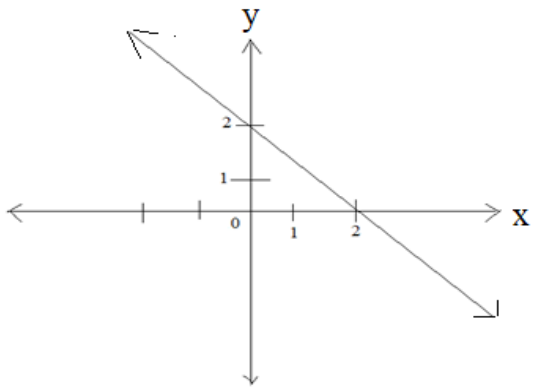
Find;

(i) $A \cap B$
(ii) $A' \cap B'$

(b) In a class of 120 students, 40 study English, 60 study Kiswahili and 30 study both Kiswahili and English. Use Venn Diagrams to find the number of students who study: -

(i) English Only
(ii) Neither English nor Kiswahili

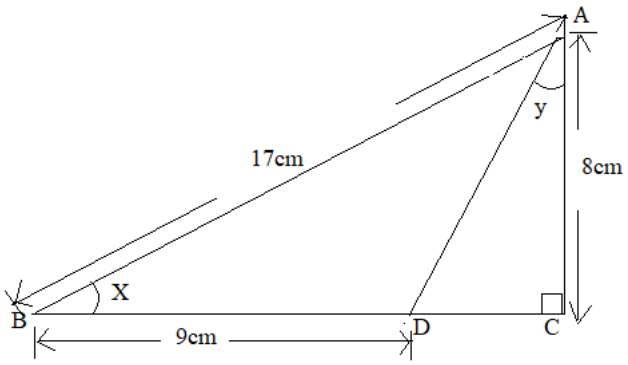
9. (a) With the help of graph, find the coordinate of a point at which a line through (-1,3) with a slope of -1, intersect a line through (5,3) with a slope of -2
(b) From the following graph of a straight line find;
(i) Equation of a line in the form of $ax + by + c = 0$
(ii) Slope of the line.



10. (a) (i) Without using calculators simplify $(cos^2 45^\circ - sin^2 30°) \div (cos 30° + \sin 60°)$
(ii)Use the following figure to calculate the value of $cos\ x + sin\ y$.

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(b) A man who is 172cm tall notices that his shadow measures 158cm in length. Find the angle of elevation of the sun.

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